

Design Method of Morphological Structural Function for Pattern Recognition of EEG Signals During Motor Imagery and Cognition

Tomonari Yamaguchi, Mitsuhiro Fujio and Katsuhiro Inoue
Kyushu Institute of Technology,
 680-4, Kawazu, Iizuka, Fukuoka, Japan, 820-8502,
t_yamagu@kiri.ces.kyutech.ac.jp,
{fujio, inoue}@ces.kyutech.ac.jp

Gert Pfurtscheller
Laboratory of Brain-Computer Interface,
Graz University of Technology, Graz, Austria
pfurtscheller@tugraz.at

Abstract—Electroencephalograph (EEG) recordings during right and left hand motor imagery can be used to move a cursor to a target on a computer screen (such system is called BCI). Recently, we have proposed the detection method of Error Potential in order to add the fail safe function to BCI system. In this paper, feature extraction method based on morphological multi-resolution analysis is introduced to extract features concerned with motor imagery and cognition simultaneously from the EEG signals. Morphological filter is composed of nonlinear operation between signal and structural function and this multi-resolution analysis can be constructed by repeating this filtering to signal while changing structural function. This method is a kind of discrete wavelet analysis with non-linear characteristics and is effective to extract specific shapes. The structural function which decides the filter characteristic is designed to obtain optimal separation based on mutual information algorithm or spectrum dividing algorithm.

Keywords—Mathematical Morphology; Multiresolution Analysis; EEG; BCI; Error Potential

I. INTRODUCTION

It is known that Electroencephalograph (EEG) signal shows specific response in the event (e.g. visual stimulus, cognition and motor imagery). This characteristic can be confirmed from averaged signal. Especially, when classifying single-trial EEG signal (i.e. not averaged), it is used to control an electronic device. Such a system which transforms signals from the brain into control signals is known as a BCI [6]. For the single-trial signal recognition method, many methods have been proposed. For example, artificial neural network (ANN) with frequency components as a feature vectors and linear discrimination method based on adaptive AR parameter [3]. We have also studied a system based on AR model to investigate the learning effects of subjects [2]. However, it seems to be impossible to construct a complete BCI system, since it depends on human state. Therefore, some compensation function is needed in BCI system. The research about error potential by Schalk et al. [4] is useful to construct such function. Then, we proposed the fail safe function to be able to use the BCI system more safely [1].

In extracting the component concerned with cognition

and motor imagery, these components appear in various frequency bands and these dominant bands usually depend on subject. Therefore, these EEG signals should be decomposed by using multiresolution analysis (MRA) in order to identify significant frequency bands. In this paper, the morphological multiresolution analysis which is a kind of discrete wavelet analysis with non-linear characteristics is selected to extract these features from EEG signal. In particular, P300 of recognizing event related potential (ERP), and high frequency component of motor imagery are extracted by this method. The structural function which decide the filter characteristic is designed to obtain optimal separation based on mutual information algorithm or spectrum dividing algorithm.

II. MATERIALS AND METHODS

The Pseudo Feedback Experiment is executed to obtain EEG data during misoperation of BCI system. Left and right hand motor imagery task causes the response EEG signal caused according to hand motor imagination. Furthermore, the response EEG signal related to cognition is obtained by rare opposite feedback stimulus. In this experiment, three subjects (N, U, Y: 20-24 years old) participated for three days. Subject (N) and (U) was trained up enough for this experimental system. Subject (Y) did not have training, but he was well informed of the experimental situation and system. The personal specification and learning effect of subjects is confirmed by this experiment. During the experiment, subject fixated a computer monitor 100 cm in front of him. Each trial was 8000 ms long (Fig. 1). At 3000 ms, the fixation cross was overlaid with an arrow at the center of the monitor for 1250 ms, pointing either to the left or to the right. Depending on the direction of the arrow, the subject was instructed to imagine a movement of the left hand or the right hand. Feedback (FB) consisting of a bar-graph is presented at the center of the monitor from 5000 ms to 8000 ms. This bar is presented same orientation of motor imagery in frequent (correct class). On the other hand, the bar shows opposite orientation in rare (error class). This ratio is set to 8:2 in this experiment. The EEG signals are sampled at 512 Hz.

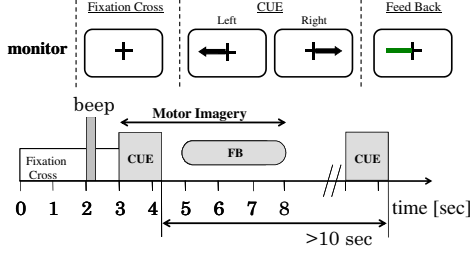


Figure 1. Timing chart

III. FEATURE EXTRACTION METHOD

A. Morphology

In this section, we give a brief description on mathematical morphology. As basic operations, we employ Minkowski addition $\oplus$ and Minkowski subtraction $\ominus$, which are defined as follows.

$$[f \oplus g](t) = \max_{\substack{t-u \in F \\ u \in G}} \{f(t-u) + g(u)\} \quad (1)$$

$$[f \ominus g](t) = \min_{\substack{t-u \in F \\ u \in G}} \{f(t-u) - g(u)\} \quad (2)$$

where $f(t)$ is an original signal and $g(t)$ is the structural function which characterizes the filter. Furthermore, F and G denote the domains of $f(t)$ and $g(t)$, respectively. Conventionally, it is assumed that every signal takes the value $-\infty$ out of its domain. By combining these operations, we define two morphological filters:

$$\text{opening} : f_g(t) = [(f \ominus g^s) \oplus g](t) \quad (3)$$

$$\text{closing} : f^g(t) = [(f \oplus g^s) \ominus g](t) \quad (4)$$

where $g^s(t) := g(-t)$. The opening process for $f(t)$ by $g(t)$ removes the parts in the positive direction of the wave form of $f(t)$ those are too narrow to fit for the wave form of $g(t)$ attached below. In contrast, the closing filter removes the narrow negatively directed parts. In other words, opening (resp. closing) smooth $f(t)$ from the positive (resp. negative) direction by $g(t)$. Furthermore, the open-closing filter consisting of successive applications of the opening followed by the closing provides an effect of lowpass filter. Thus, we can also construct a highpass filter by taking the difference between the original signal and its open-closing.

$$\text{lowpass} : \psi^\uparrow(t) = (f_g^s)^\uparrow(t) \quad (5)$$

$$\text{highpass} : \omega^\uparrow(t) = f(t) - \psi^\uparrow(t) \quad (6)$$

The schematic figures of morphological operations are shown in Fig. 2.

A sequence of successive processes with varying structural functions constitutes a multiresolution signal analysis. To describe this, let us assume that there exist sets V_j at level j . Then, for a given input signal $x^0 \in V_0$, we obtain the following recursive analysis scheme:

$$x^0 \rightarrow \{x_L^1, x_H^1\} \rightarrow \{x_{LL}^2, x_{LH}^2, x_{HL}^2, x_{HH}^2\}$$

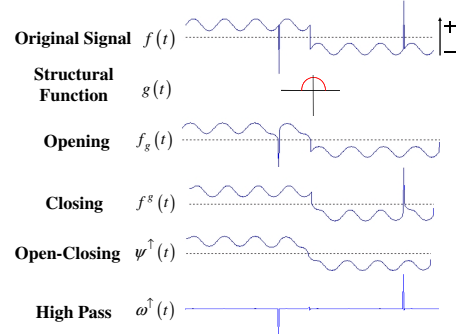


Figure 2. Property of morphological filter

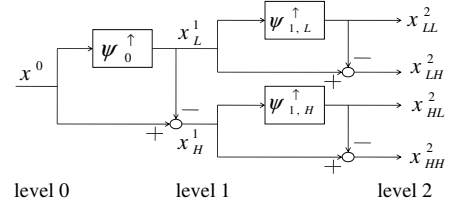


Figure 3. Signal analysis scheme

Conversely, the original signal x_0 can be reconstructed by summing up the signals. A three-level signal analysis scheme is depicted in Fig. 3.

B. Optimization of structural function

In processing mathematical morphology, the characteristics of filter is decided by the shape of structural function. Some parametric structural functions such as Ellipse, Gaussian and Cosine are defined [5]. In this paper, the shape of the structural function is designed according to ellipse (Fig. 4) because this shape takes very simple structure. The ellipse type is composed of two parameters (n of the time axis parameter and k of the value axis). This parameters are optimized to become different during separated signals x_L and x_H by random search. The condition of difference is evaluated by using two methods as follows.

1) Mutual information

The heuristic parameters of structural function are optimized by using mutual information between separated signals $\mathbf{x}^j = \{x_L^j, x_H^j\}$. The evaluation function is the joint entropy between x_L^j and x_H^j .

$$H(\mathbf{x}^j) = \frac{1}{2} \log\{(2\pi e)^2 |\det(\Sigma)|\} \quad (7)$$

where, Σ is covariance matrix of $\mathbf{x}^j$.

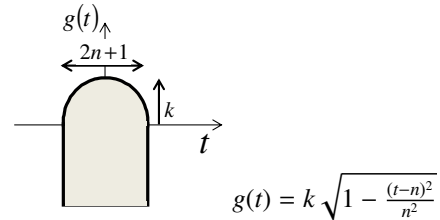


Figure 4. Model of structural function

2) Spectrum dividing

The heuristic parameters are optimized to become separating between signals in frequency domain. In this algorithm, the prior parameter of separation frequency F_{sp} , the penalty coefficient p and prior parameters of evaluation interval F_{min} and F_{max} need to decide before processing.

$$C(\mathbf{x}^j) = \frac{1}{F_{sp} - F_{min}} \int_{F_{min}}^{F_{sp}} (\hat{x}_L^j(f) - p \cdot \hat{x}_H^j(f)) \cdot \hat{x}^{j-1}(f) df + \frac{1}{F_{max} - F_{sp}} \int_{F_{sp}}^{F_{max}} (\hat{x}_H^j(f) - p \cdot \hat{x}_L^j(f)) \cdot \hat{x}^{j-1}(f) df \quad (8)$$

where, x^{j-1} is original signal of $\mathbf{x}^j$ and $\hat{\mathbf{x}}^j = \{\hat{x}_L^j, \hat{x}_H^j\}$ is Fourier spectrum of $\mathbf{x}^j$.

C. Pattern recognition methods

It is thought that positive potential P300 concerned with cognition appears in error class. Therefore, decision rule of correct class and error class is composed of amount of positive component as follows:

$$k^* = \text{ArgMax}_k \Pr(C_k | s) \quad (9)$$

$$= \text{ArgMin}_k \left\{ \ln(\sigma_k) + \frac{1}{2\sigma_k^2} (s - m_k)^2 - \Pr(C_k) \right\}$$

where,

$$s = \sum_{t=1}^N \{H(\tilde{y}(t)) \cdot \tilde{y}(t)\}, H(\tilde{y}) = \begin{cases} 1, \tilde{y}(t) \geq 0 \\ 0, \tilde{y}(t) < 0 \end{cases}$$

also, m_k is mean and σ_k is variance of s . The phenomenon of event-related synchronization and desynchronization occur before and after motor imagery. The decision rule of left hand motor class and right hand motor class is constructed under the assumption that distribution of EEG amplitude is normal distribution in each class as follows:

$$k^* = \text{ArgMax}_k \Pr(\omega_k | \tilde{y}) = \text{ArgMin}_k \left[\frac{1}{2} \sum_{t=1}^N \ln \sigma_{k,t}^2 + \frac{1}{2} \sum_{t=1}^N \left\{ \frac{1}{\sigma_{k,t}^2} (\tilde{y}(t) - \bar{\tilde{y}})^2 \right\} - \ln \Pr(\omega_k) \right] \quad (10)$$

$$\bar{\tilde{y}}_k(t) = \sum_{j=1}^M \tilde{y}_k^j(t) \quad (M : \text{block num.})$$

$$\sigma_{k,t}^2 = \frac{1}{2} \sum_{j=1}^M (y_k^j(t) - \bar{\tilde{y}}_k(t))^2$$

IV. RESULTS

A. Simulation

The proposed algorithm was applied to synthesized signal. This signal is composed of two kinds sinusoidal wave (5 Hz and 20 Hz) and white noise. Structural function of Ellipse type which composed of two heuristic parameters is

designed by using two kinds optimization algorithm in Eqs. (7) and (8). The suitable heuristic parameters are obtained by the random search about 10000 times. One result of mutual information algorithm is depicted in Fig. 5. In this case, the SNR is 7.98 and each component is separated respectively. In separation of spectrum dividing algorithm, same components are extracted. In addition, spectrum dividing algorithm seems to have robustness with noise (Table I). This algorithm is able to obtain the effectiveness component from the noisy signal than mutual information algorithm (Fig. 6). Though, mutual information algorithm need not be decide prior parameters such as F_{min} , F_{max} , F_{sp} and p in Eq. 8. Therefore, mutual information algorithm is suitable for separation of the practical signal which is not understood components.

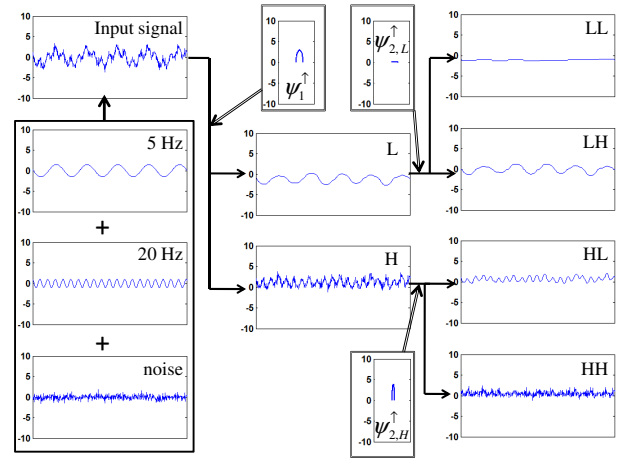


Figure 5. Analysis of less-noisy synthesized sinusoidal wave (algorithm: Mutual Info., SNR: 7.98, repeat count: 10000)

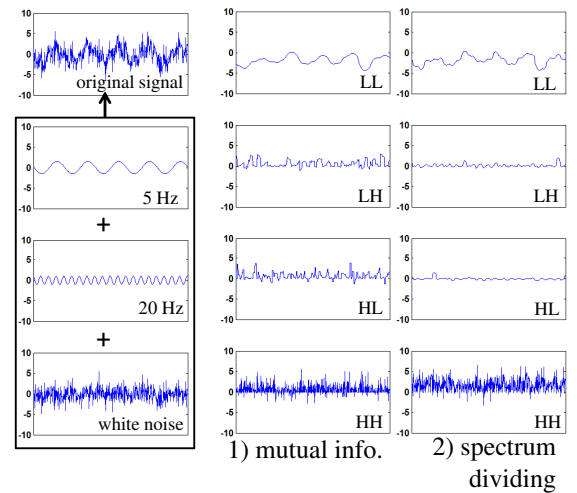


Figure 6. Analysis of noisy synthesized sinusoidal wave (SNR: -1.57, SR: 500 Hz, repeat count: 10000, $F_{min} = 2$ Hz, $F_{max} = 30$ Hz, $F_{sp} = 23, 17, 30$ Hz, $p = 0.2$)

Table I
COEFFICIENT OF CORRELATION COMPARED WITH ORIGINAL SIGNALS

algorithm	MI	SD	MI	SD
SNR	7.98	7.98	-1.57	-1.57
5Hz	0.95	0.94	0.83	0.83
20Hz	0.79	0.82	0.59	0.58
noise	0.82	0.83	0.77	0.89

(MI: Mutual Information. SD: Spectrum Dividing)

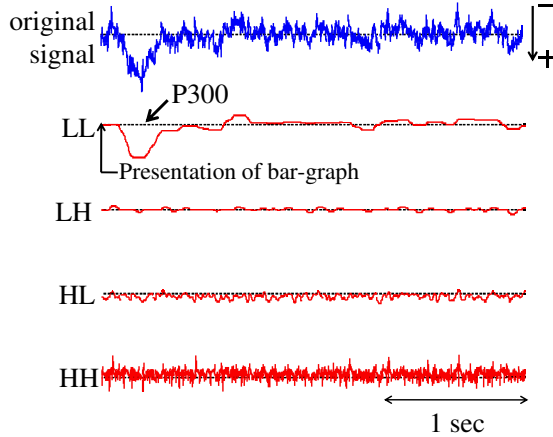


Figure 7. EEG analysis by using morphology
(subject: N, session no: 1, CH: C4, data: 5.0 ~ 8.0 sec
Algorithm: Mutual Information)

B. Experimental Result

EEG signals is separated by using mutual information algorithm. The data period for analysis is selected from 5000 msec to 8000 msec and the heuristic parameter is obtained by random search about 1000. Four levels of morphological multiresolution analysis processes to single trial EEG signal. At the time just after presentation of bar-graph, the peak in positive direction is observed at a specific level, it is P300 (Fig. 7). The classification results concerned with motor imagery (left and right motor imagery class) and error potential detection(correct and error class) for each level are shown in Figs. 8 and 9. In subject N, it was confirmed that accuracy rose about 10 % in this situation. In the other subject Y, it was improved only in the case of error potential detection. These reasons are thought that the morphology method is able to extract high amplitude and high frequency components concerned with motor imagery and P300-potentials concerned with cognition.

V. CONCLUSION

In this paper, feature extraction based on morphological filter was introduced to discriminate the EEG signals recorded during hand motor imagery and cognition. The structural function to decide the filter characteristic was selected to obtain optimal separation by selecting the feature to mutual information and spectrum dividing algorithm. It was confirmed to the method using spectrum dividing algorithm was effective for the signal which mixed a lot of noise. Though, from the viewpoint of prior parameter

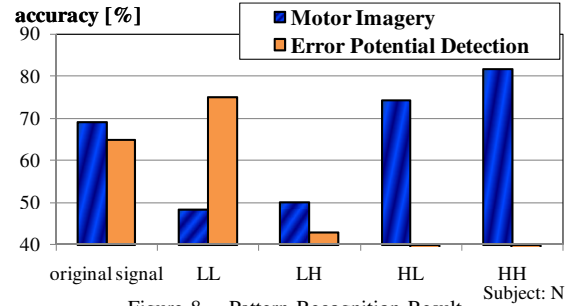


Figure 8. Pattern Recognition Result
(subject: N, session no: 1, CH: C4, data: 5.0 ~ 5.5 sec
Algorithm: Mutual Information)

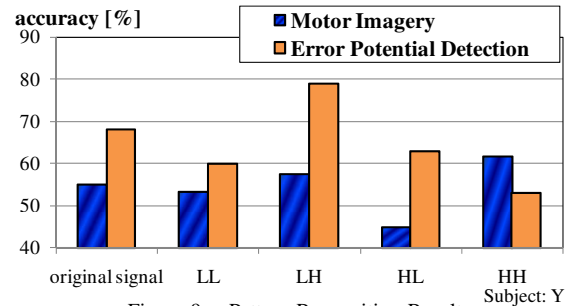


Figure 9. Pattern Recognition Result
(subject: Y, session no: 2, CH: C4, data: 5.0 ~ 5.5 sec
Algorithm: Mutual Information)

setting, the mutual information algorithm is suitable for processing of EEG signals. Consequently, we confirmed that features of motor imagery and Error Potential in EEG signal is extracted effectively to each band. A BCI system with the fail safe function will be able to be achieved by advancing this research further.

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